Week 10: November 2-6
Chemistry in Biology

Subject/:الموضوع Biology **Unit Title: عنوان الوحدة The Cell** **Gradeالصف : 9**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1	I can explain and classify the properties of water and aqueous mixtures by modelling water polarity and hydrogen bonding, distinguishing solutions from suspensions and colloids, and interpreting pH and buffer functions.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Water and Its Solutions
Lesson 2	I can explain how scientific discoveries established cell theory by sequencing historical discoveries, comparing microscope technologies, calculating magnification, and distinguishing prokaryotic from eukaryotic cells.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Cell Discovery and Theory
Lesson 3	I can explain how cell structures support cellular functions by identifying organelles, matching each organelle to its function, and tracing how several organelles work together to produce and transport proteins.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Structures and Organelles
Lesson 4	I can compare plant and animal cells and explain how their structures support cellular functions by identifying shared and specialized organelles, analyzing cell movement structures, and modelling how organelles cooperate in protein production and transport.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- Structures and Organelles
Homework الواجبات			

Teacher's notes/ Reminders ملاحظات المعلم والتذكير	

#الأسبوع Week # 10

Technology Integration دمج التكنولوجيا	digital molecule models to identify the polar regions of water digital or projected images of solutions, suspensions, and colloids help students compare mixtures historical cell images and digital micrographs interactive plant and animal cell models to identify organelles and reveal their functions
Cross-curricular links الروابط بين المواد الدراسية	Chemistry: Molecular polarity, intermolecular attractions, solutions, acids, bases, and pH. Mathematics: Ordering pH values and interpreting numerical scales. History: The chronological development of cell discovery and scientific ideas. Chemistry: Protein production, chemical reactions, and energy conversion. Mathematics: Organizing organelle-function data in comparison tables and interpreting diagrams.
External Assessment Integration دمج التقييمات الخارجية	NA
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Research & Thinking (ATL): Classify samples using observable evidence, identify patterns in pH values, and justify conclusions. Communication: Use scientific vocabulary to distinguish polarity, intermolecular attractions, solutions, suspensions, and buffers. Collaboration: Compare observations, agree on classifications, and provide evidence for group decisions. Research & Thinking (ATL): Evaluate evidence, distinguish observations from conclusions, and explain how new technology changes scientific understanding. Communication: Present historical discoveries and cell theory principles using concise scientific explanations. Collaboration: Build a shared timeline and compare interpretations of microscope images. Research & Thinking (ATL): Predict the effects of organelle malfunction and explain how evidence supports a structure–function relationship. Communication: Use labelled diagrams and accurate vocabulary to explain organelle functions and pathways. Collaboration: Compare organelle models, correct pathway sequences, and provide constructive feedback.

Standards المعايير	HS-LS1-6: Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How do scientists use models of molecules and measurements such as pH to explain properties that cannot be understood through observation alone? How does the development of new technology change what scientists can claim to know about the natural world? To what extent can understanding the individual parts of a cell explain how the entire cell functions?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 MUST DOK 1 – Recall: Define polarity, solution, solvent, solute, acid, base, pH, and buffer. SHOULD DOK 2/3 – Apply & Analyze: Classify mixtures and pH samples, using evidence to explain the differences between solutions and suspensions and the role of buffers. COULD DOK 4 – Evaluate & Create: Design and justify an investigation comparing mixtures in water, including controlled variables and observations that would support each classification. Lesson 2 MUST DOK 1 – Recall: State the three principles of cell theory and identify key scientists associated with cell discovery. SHOULD DOK 2/3 – Apply & Analyze: Calculate total magnification and compare microscope technologies and cell types using appropriate evidence. COULD DOK 4 – Evaluate & Create: Create and justify a historical evidence timeline showing how discoveries and microscope improvements contributed to cell theory. Lesson 3 MUST DOK 1 – Recall: Identify major organelles and state their functions. SHOULD DOK 2/3 – Apply & Analyze: Trace the pathway of a protein through the rough ER, Golgi apparatus, and vesicles and explain the role of each structure. COULD DOK 4 – Evaluate & Create: Construct and justify an integrated cell model showing how organelles cooperate, including the likely consequences of disrupting one component. Lesson 4

	MUST DOK 1 – Recall: Identify shared and distinguishing structures in plant and animal cells and state their functions. SHOULD DOK 2/3 – Apply & Analyze: Compare plant and animal cells and explain how the plasma membrane, cell wall, cilia, flagella, and organelles contribute to cellular functions. COULD DOK 4 – Evaluate & Create: Construct and justify a comparative plant and animal cell model showing structure–function relationships and predicting the consequences of disrupting one organelle.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Citizenship — Conservation Values – Global understanding
	Area: Responsible water use, water quality, and environmental conservation in the UAE. Area: Scientific discovery, technological innovation, and the contribution of international scientific collaboration to understanding living organisms. Area: Scientific literacy, biotechnology, and understanding how cellular processes support living organisms.
	How to Implement: Use examples of freshwater and seawater in the UAE to discuss why water quality matters for people and ecosystems. Students interpret sample pH values and propose responsible practices for protecting water resources. Emphasize that pH is one indicator of water quality and cannot, on its own, establish whether water is safe to drink. How to Implement: Students examine how scientists from different periods and backgrounds contributed to cell theory. Discuss how scientific progress depends on sharing evidence, improving technology, and revising explanations. Connect these principles to the value of scientific research and innovation in the UAE. How to Implement: Students investigate how understanding organelle function contributes to research in areas such as plant production, health, and biotechnology. They discuss how scientific knowledge can support responsible innovation and help address biological challenges relevant to the UAE and the wider world.